Report on

The target-free clique conjecture for threshold-linear networks: complete exact maps for $n \leq 7$, the $\varepsilon \to 0$ map at $n = 8$, infinite families, and unbounded thresholds

This report is based on the supplied 33-page draft dated October 2026. The repository materials, including code, certificates and deferred proof notes, were not reviewed.

Potentially important issues

1. Lower-dimensional cells can contain stable fixed points (Section 3.1, page 8). The final sentence before Section 3.2 says that lower-dimensional cells are boundary points and are never SFP. In the cylindrical algebraic decomposition described there, vertical sections above roots of projection polynomials need not lie on any of the original boundary curves. They can pass through the open SFP region.

For a concrete example, take G_1 from Table 1 and write $\Delta_5(\beta, t)$ for the fifth Hurwitz determinant of $\det(\mu I + K(\beta, t))$. There is exactly one root $\beta_0 \in (17/8, 15/7)$ of

$$\Delta_5\left(\beta, \frac{1}{\beta - 2}\right) = 0.$$

This is a projection value arising from the resultant with the legality boundary $t(\beta - 2) - 1 = 0$. Nevertheless, at $(\beta_0, 100)$ all coordinates of $\mathrm{adj}(I + \beta_0 N)\mathbf{1}$ and all six Hurwitz determinants are strictly positive, and $100(\beta_0 - 2) > 1$. Thus the full support is an SFP on a lower-dimensional vertical section.

These positivity inequalities hold throughout $\beta \in [17/8, 15/7]$ at $t = 100$. An exact check is to substitute $\beta = (17/8 + (15/7)u)/(1 + u)$ into each of the six adjugate coordinates and six Hurwitz determinants and multiply by $(1+u)$ to the degree of that polynomial: every coefficient is positive. The unique root above is obtained by exact root counting in the stated interval.

The issue concerns the claimed pointwise description of the regions. Openness still makes full-dimensional cells sufficient for detecting the existence of an SFP, so this example does not disprove the census counts or threshold values. The treatment of these sections in the implementation and recorded regions was not reviewed.

Very minor points

2. Theorem 5.4 does not cover the endpoint used in Lemma 7.4 (page 26). The proof begins by invoking Theorem 5.4 with $\nu_\infty = 1/2$, whereas that theorem explicitly assumes $\nu_\infty \in (1/2, 1)$. The theorem as stated therefore does not supply the expansion used here. This is a local justification gap; it does not refute Lemma 7.4 or Theorem 7.5.

3. The primitive proportions decrease (Section 6.6, page 23). The paragraph after Table 6 says that the share of primitive counterexamples grows from 3/3 to 102/123 to 6720/9133. These proportions satisfy $1 > 102/123 = 34/41 > 6720/9133$ and hence decrease; the absolute counts grow.